

DBSCAN Clustering

① K-Means Limitation

② DBSCAN Procedure

Working

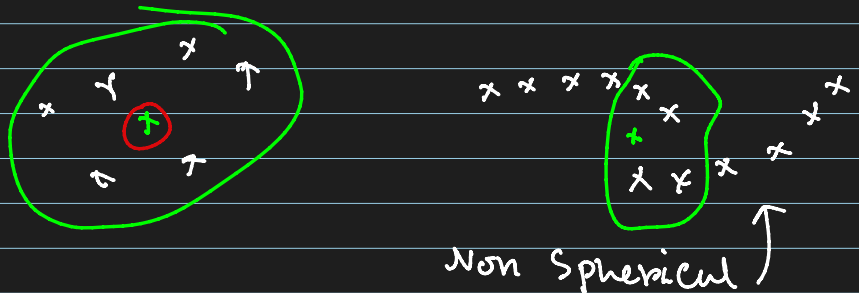
core points, Border,
noise points

Procedure

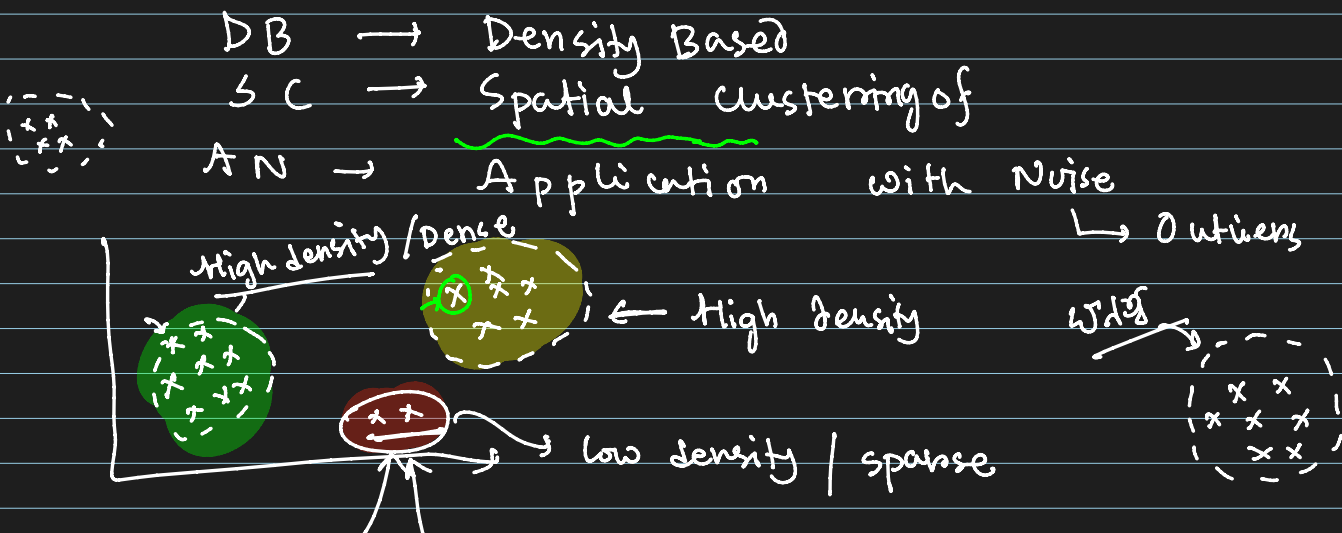
Code.

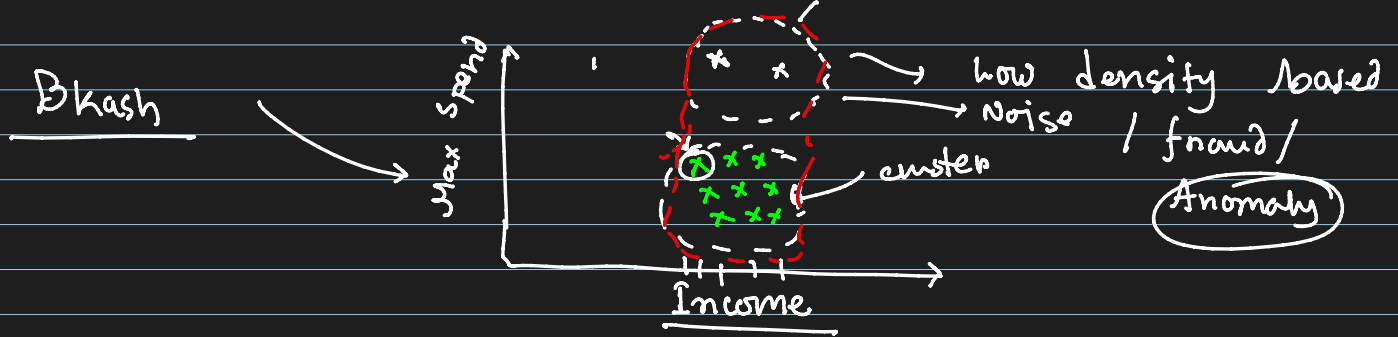
③ K-Means comparison.

Limitations of K-Means Clustering

- ① specify the cluster numbers → ① feat ↑
 - ✓ ① Outliers Sensitive. → Centroid Based
 - ① Works well only with spherical
- 
- Non Spherical

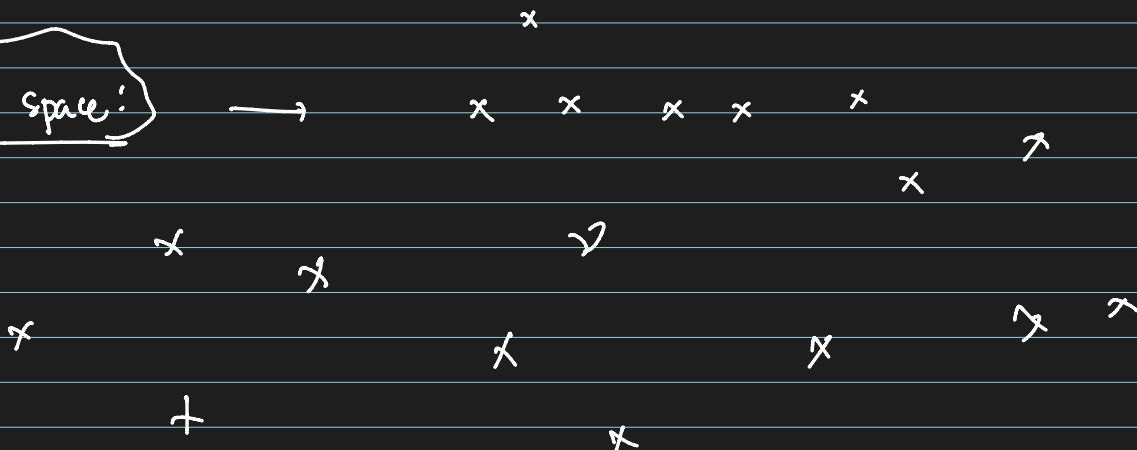
- DBSCAN Clustering → ① No need to specify clusters
- Density based.
 - ① Robust to outliers
 - ① Works well with spherical & Non spherical



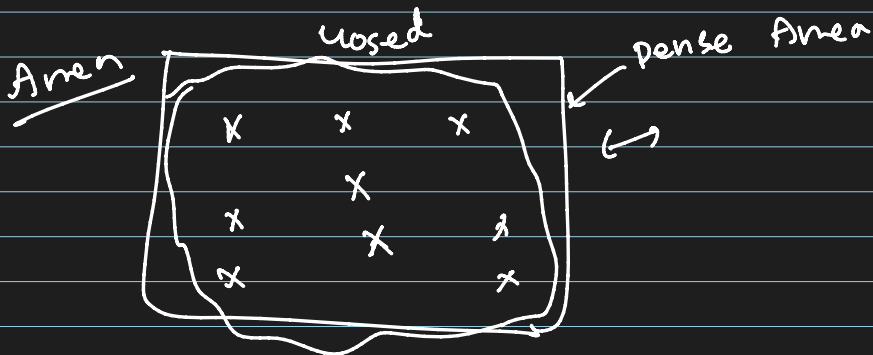


Two important factors of DBSCAN

Open space:



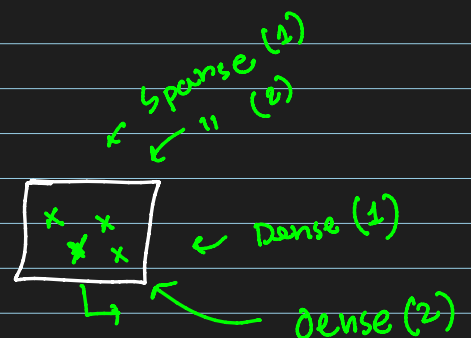
clean idea



clean idea → Area (closed)
→ Number of entity
↑
Dense

Ex: Min Number of entity = 3

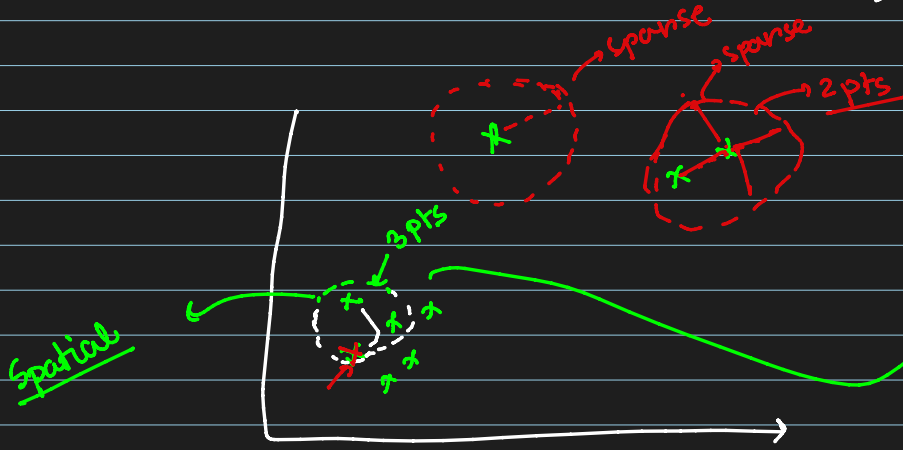
Area → $2m$



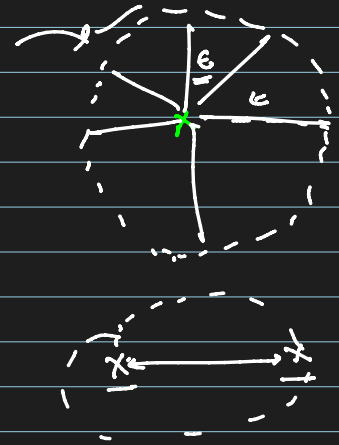
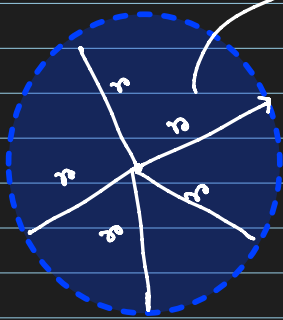
DBSCAN hyperparameters

epsilon (ϵ) → Area

Min-pts → Number of entity.



Area → Circle → ϵ → radius



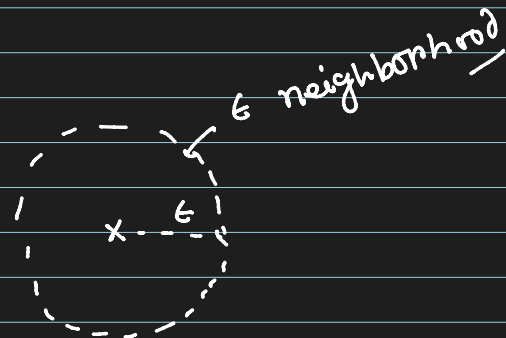
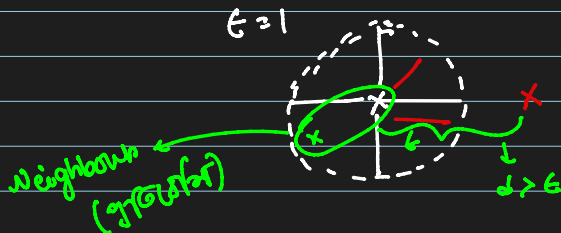
xxx

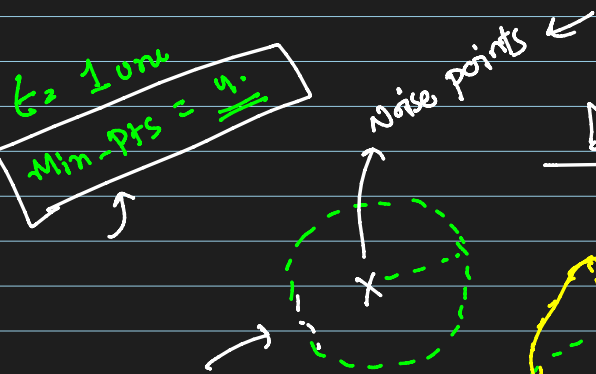
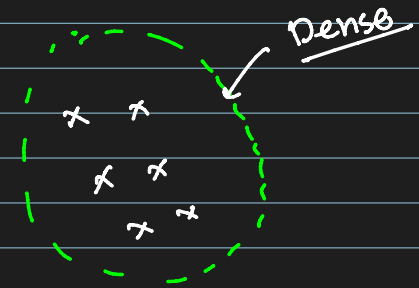
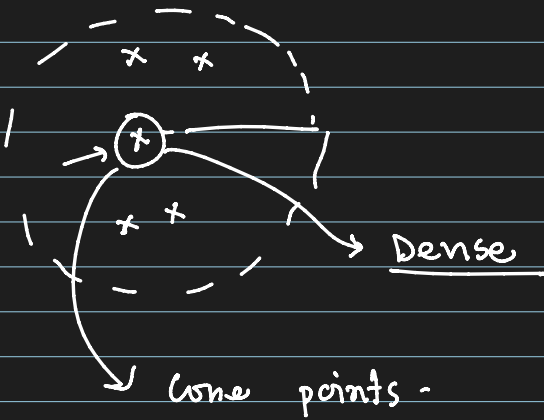
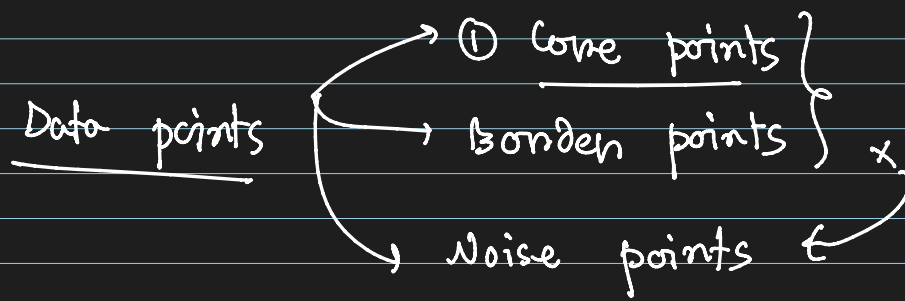
Epsilon

(ϵ) → maximum distance within which two points are considered neighbour.

Min-pts

The number of points/sample to consider to call a point, core point in its ϵ neighbourhood.

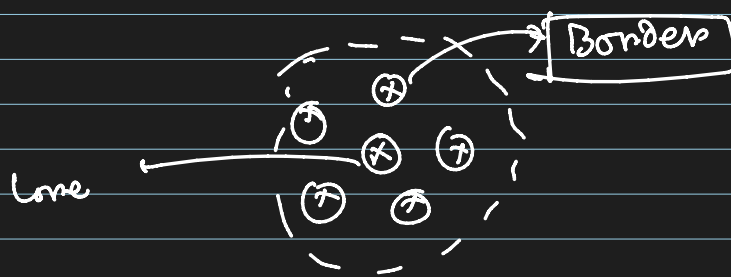
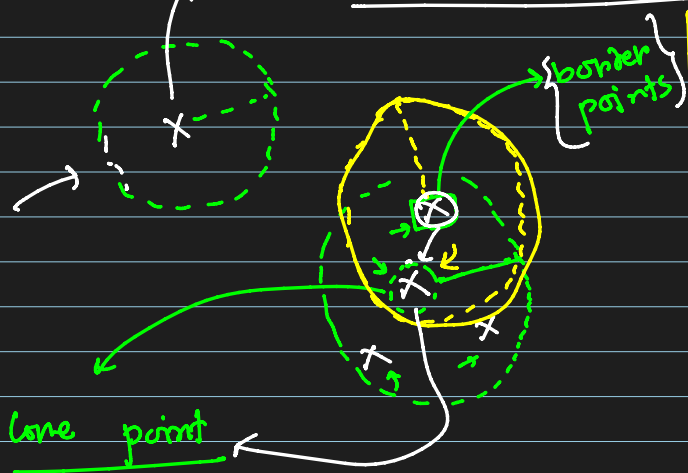




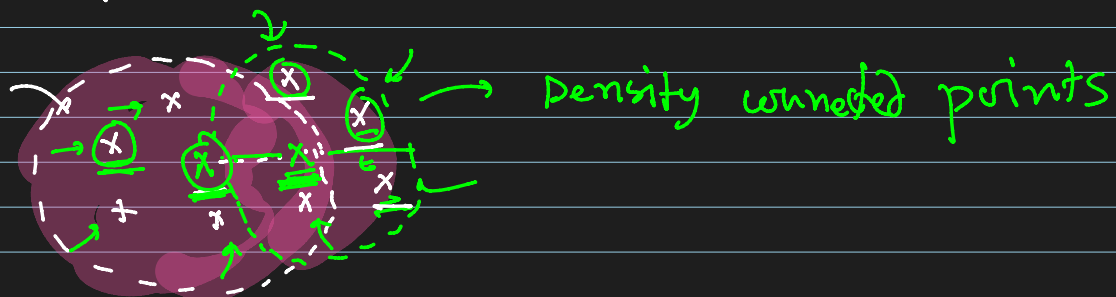
Border Points

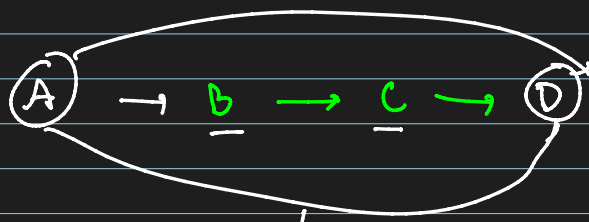
Core point at
 ϵ neighborhood
 4 min-pts around
 pts not

ϵ -neighborhood 4 min
 4 not core points (ATZ)

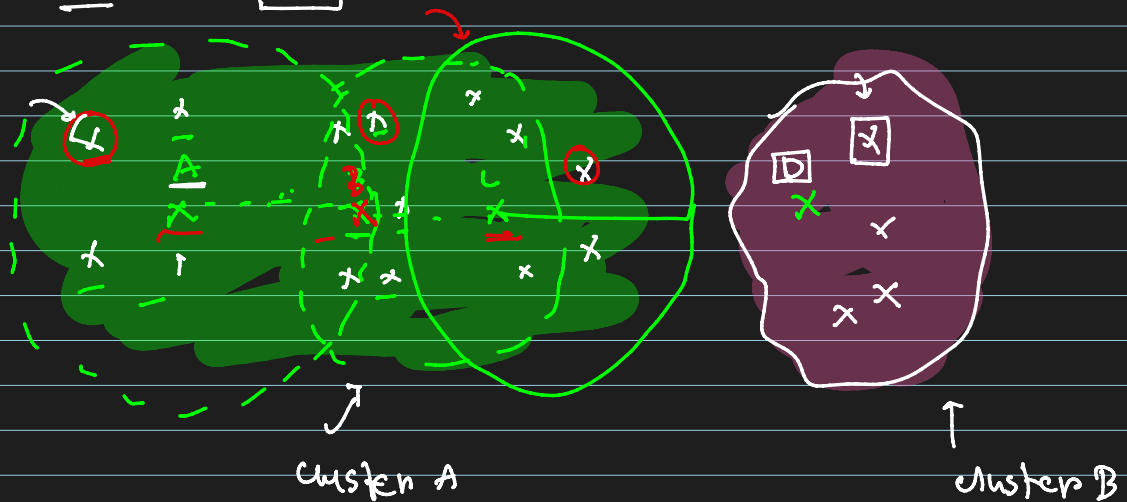
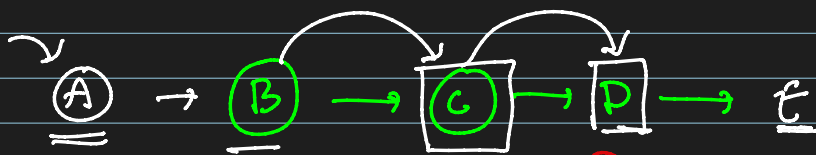


① Core points has the capability to make a cluster and to expand the cluster.





Density connected points



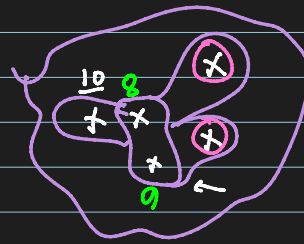
Working Procedure of DBSCAN

- ① $\epsilon = 1$, min-pts = 3.
- ② Identify core points, border points, noise points
- ③ Clustering the core points.
 - ↳ Based on ϵ neighborhood.
- ④ Clustering the border points.
 - ↳ to its nearest core points.

A $\rightarrow$ 1, 2, 3, 4, 5, 6, 7



Noise
x



B = 8, 9, 10

x
noise

fig: Data points.

let's implement DBSCAN

↳ Non Spherical data.

feat $\rightarrow$ 3 $\rightarrow$ min-pts (4-6)

feat $\rightarrow$ 2 $\rightarrow$ (3 $\rightarrow$ 5, 6) ✓

eps $\rightarrow$ k distance $\rightarrow$ Method

↳ elbow method

